

Lens Changes After Proton Beam Irradiation for Uveal Melanoma

EVANGELOS S. GRAGOUDAS, M.D., KATHLEEN M. EGAN, M.P.H.,
SUSAN M. WALSH, B.S., SUSAN REGAN, PH.D.,
JOHN E. MUNZENRIDER, M.D., AND VICTOR TARATUTA, PH.D.

• **PURPOSE:** Because limited data exist on the incidence of lens changes after therapeutic intraocular irradiation, we studied a group of patients who underwent proton irradiation for uveal melanoma, in order to estimate cumulative rates of radiation-induced posterior subcapsular opacities and age-related lens changes at specific time points after irradiation.

• **METHODS:** Cumulative rates for each type of opacity were estimated from among the 383 patients, treated between 1987 and 1989, who had clear lenses or minimal lens changes before irradiation. Cox's proportional hazards model was used to evaluate the independent effects of tumor and treatment-related characteristics on the development of posterior subcapsular opacities in these patients. Risk factors for posterior subcapsular opacities were evaluated.

• **RESULTS:** By three years after irradiation, posterior subcapsular opacities had developed in 42% of the patients, and rates increased significantly with lens dose and with tumor height. The risk of posterior subcapsular opacities was over three times higher when the lens received a substantial dose of irradiation (adjusted relative risk, 3.25; 95% confidence interval, 1.60–6.59), as com-

pared with minimal dose, and was three times higher when the tumor was highly elevated (adjusted relative risk, 3.05; 95% confidence interval, 1.45–6.40) as compared with minimally elevated lesions. Opacities in the other segments of the lens were age related.

• **CONCLUSIONS:** These data show that patients receiving therapeutic intraocular irradiation have a high risk of developing posterior subcapsular opacities. As expected, the leading risk determinant of posterior subcapsular change is the amount of irradiation received by the lens.

CHARGED-PARTICLE IRRADIATION IN THE TREATMENT of uveal melanoma is associated with excellent local control¹ and with retention of useful vision for long intervals in many patients.² Nevertheless, complications that threaten vision or necessitate further treatment are not uncommon in these patients; the most important of these are radiation retinopathy³ and papillopathy, neovascular glaucoma,² and cataract.

The causal role of ionizing irradiation in the development of cataract has been recognized since late in the last century. Of the various cataract subtypes (nuclear, cortical, and posterior subcapsular), only posterior subcapsular changes have been linked to microwave and ionizing radiation exposure. Therapeutic ionizing irradiation is a well-established cause of posterior subcapsular cataract,⁴ and the high susceptibility of the lens to even low levels of exposure is well known.⁵ However, relatively few data exist on the rate of lens opacification after therapeutic intraocular irradiation.

We studied the development of posterior subcapsu-

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From the Retina Service, Department of Ophthalmology, Massachusetts Eye and Ear Infirmary, Harvard Medical School (Drs. Gragoudas and Regan, Ms. Egan, and Ms. Walsh), the Department of Radiation Medicine, Massachusetts General Hospital, Harvard Medical School, Boston, Massachusetts (Dr. Munzenrider); and the Oculon Corporation, Cambridge, Massachusetts (Dr. Taratuta). This study was supported in part by a grant from Oculon Corporation, Cambridge, Massachusetts, and by the Uveal Melanoma Fund of the Massachusetts Eye and Ear Infirmary, Boston, Massachusetts.

Reprint requests to Evangelos S. Gragoudas, M.D., Massachusetts Eye and Ear Infirmary, 243 Charles St., Boston, MA 02114.